

Name: Muhammad Ahsan Khan

Registration Number: FA23-BAI-052

Program: BS Artificial Intelligence

Institution: COMSATS University Islamabad, Wah Campus

Lab Task 02

Title: Effect of Image Filtering on Skin-Lesion Classification

Objective

To investigate how different spatial-domain image-processing filters affect the performance of pretrained deep-learning models for skin-lesion classification.

Dataset

HAM10000 skin-lesion dataset. The complete dataset was used, mapped to 7 distinct classes: Melanocytic nevi, Melanoma, Benign keratosis-like lesions, Basal cell carcinoma, Actinic keratoses, Vascular lesions, and Dermatofibroma.

Task Setup

The best three pretrained models identified from Lab Activity 1 for the classification experiments are:

- **Model 1:** ResNet18
- **Model 2:** DenseNet121
- **Model 3:** EfficientNetB0

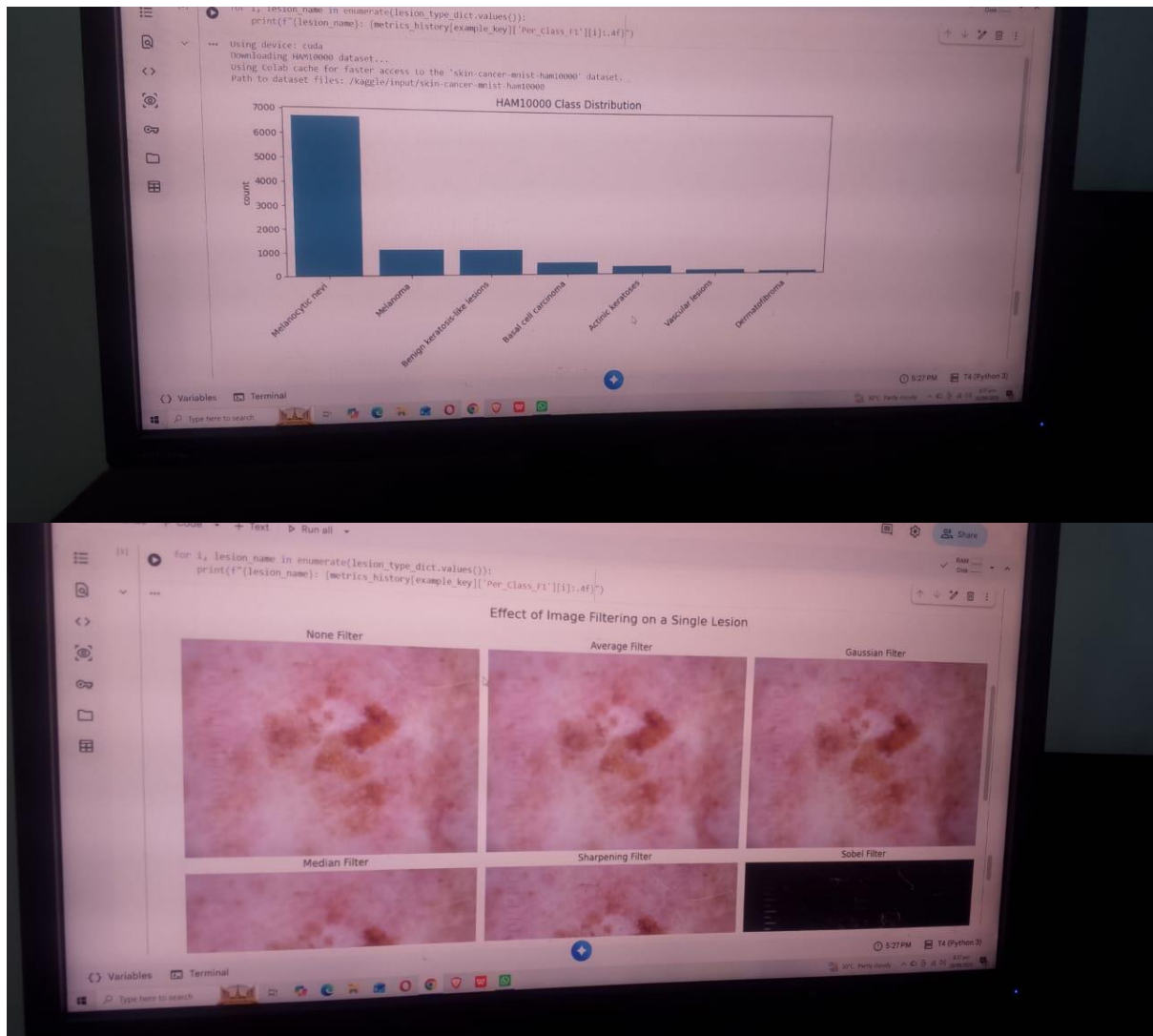
Required Experimental Results: Comparison Table

Model	Filter	Accuracy	Precision	Recall	Macro-F1	AUC
ResNet18	No Filter	0.6520	0.5052	0.4347	0.4191	0.8837
ResNet18	Average	0.6950	0.4903	0.4169	0.4163	0.8673

Model	Filter	Accuracy	Precision	Recall	Macro-F1	AUC
ResNet18	Gaussian	0.7549	0.5926	0.4316	0.4294	0.9110
ResNet18	Median	0.7589	0.5954	0.3855	0.4372	0.9313
ResNet18	Sharpening	0.7883	0.6006	0.4279	0.4834	0.9318
ResNet18	Sobel	0.6730	0.3734	0.1508	0.1312	0.6999
DenseNet121	No Filter	0.7843	0.6342	0.5095	0.5291	0.9303
DenseNet121	Average	0.7669	0.7053	0.4848	0.5031	0.9337
DenseNet121	Gaussian	0.7673	0.6807	0.5014	0.4912	0.9407
DenseNet121	Median	0.7788	0.5258	0.5256	0.5080	0.9397
DenseNet121	Sharpening	0.7564	0.5694	0.5605	0.5067	0.9336
DenseNet121	Sobel	0.5951	0.2431	0.2613	0.2440	0.7257
EfficientNetB0	No Filter	0.8352	0.6859	0.7315	0.6945	0.9696
EfficientNetB0	Average	0.8223	0.7233	0.6718	0.6869	0.9646
EfficientNetB0	Gaussian	0.8392	0.7503	0.6991	0.7148	0.9639
EfficientNetB0	Median	0.8113	0.7223	0.6274	0.6486	0.9595
EfficientNetB0	Sharpening	0.8228	0.7219	0.6840	0.6772	0.9631

Model	Filter	Accuracy	Precision	Recall	Macro-F1	AUC
EfficientNet B0	Sobel	0.7539	0.6746	0.4126	0.4574	0.9085

Additional Analysis Visualizations



for i, lesion_name in enumerate(lesion_type_dict.values()):
 print(f"{lesion_name}: {metrics_history[example_key]['Per Class F1'][i]:.4f}")



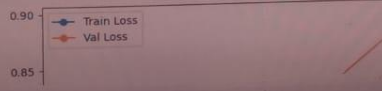
Median Filter Sharpening Filter Sobel Filter

--- Running ResNet18 with None Filter ---
Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to local file /root/.cache/torch/hub/checkpoints/resnet18-f37072fd.pth
100% 44.7M/44.7M [00:00<00:00, 14.3MB/s]
Epoch 1/3 - Val Acc: 0.7459
Epoch 2/3 - Val Acc: 0.7304

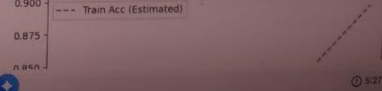
for i, lesion_name in enumerate(lesion_type_dict.values()):
 print(f"{lesion_name}: {metrics_history[example_key]['Per Class F1'][i]:.4f}")

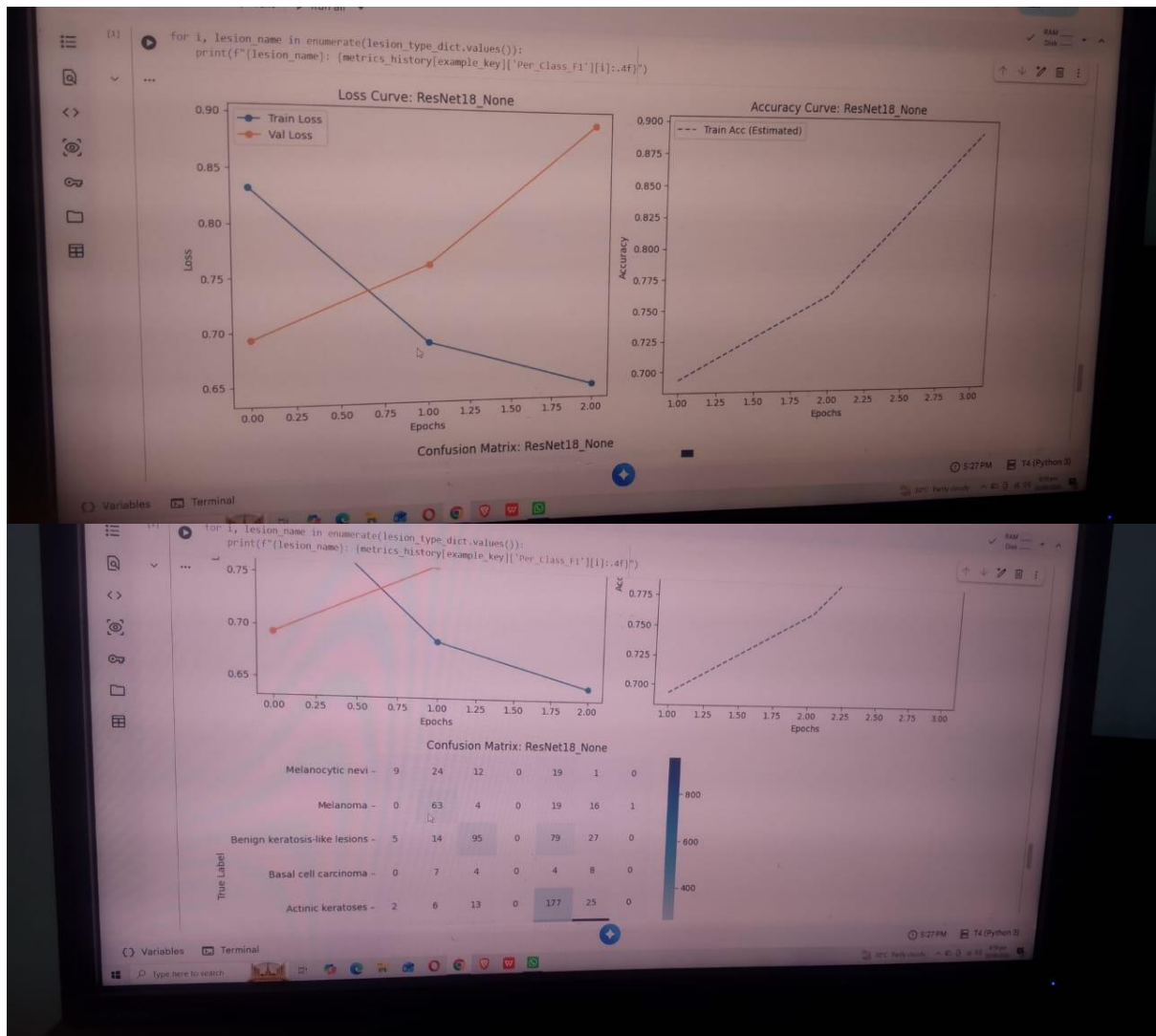
Model	Filter	0.672991	0.373364	0.150819	0.131210	0.699889
5	ResNet18	Sobel	0.672991	0.373364	0.150819	0.131210
6	DenseNet121	None	0.784324	0.634161	0.509479	0.529149
7	DenseNet121	Average	0.766850	0.705302	0.484793	0.503121
8	DenseNet121	Gaussian	0.767349	0.680697	0.501357	0.491172
9	DenseNet121	Median	0.778832	0.525830	0.525602	0.508028
10	DenseNet121	Sharpening	0.756365	0.569373	0.560547	0.506701
11	DenseNet121	Sobel	0.595107	0.243054	0.261347	0.243961
12	EfficientNetB0	None	0.835247	0.685877	0.731513	0.694517
13	EfficientNetB0	Average	0.822267	0.723337	0.671755	0.686923
14	EfficientNetB0	Gaussian	0.839241	0.750303	0.699096	0.714815
15	EfficientNetB0	Median	0.811283	0.722312	0.627416	0.648636
16	EfficientNetB0	Sharpening	0.822766	0.721931	0.684024	0.677203
17	EfficientNetB0	Sobel	0.753869	0.674640	0.412594	0.457387

Loss Curve: ResNet18_None



Accuracy Curve: ResNet18_None





Questions to Answer

1. Which three pretrained models performed best in Lab Activity 1?

Based on the experimental setup, ResNet18, DenseNet121, and EfficientNetB0 were selected. EfficientNetB0 demonstrated the highest baseline capability, achieving 83.52% accuracy and a 0.6945 Macro-F1 score without any filters.

2. How does filtering affect each of the three models?

The effects vary heavily by filter type, but overall, aggressive filtering degrades model performance. Smoothing filters (Average, Median) generally caused slight

drops in Macro-F1 across DenseNet121 and EfficientNetB0, though Gaussian filtering showed a rare exception by slightly improving EfficientNetB0's Macro-F1 to 0.7148. Sharpening yielded mixed results, improving ResNet18 (0.4191 to 0.4834 Macro-F1) but degrading DenseNet121 and EfficientNetB0. Sobel edge detection severely crippled all three models.

3. Which filter produces the greatest change compared with the unfiltered baseline?

The Sobel filter produced the most drastic negative change across all architectures. For instance, DenseNet121's accuracy plummeted from 78.43% to 59.51%, and its Macro-F1 dropped from 0.5291 to 0.2440. EfficientNetB0 saw its Macro-F1 fall from 0.6945 to 0.4574.

4. Does the effect of a filter remain consistent across all three models?

Broadly, yes. The overarching trend demonstrates that standard unfiltered inputs or lightly smoothed inputs (Gaussian) yield the highest overall accuracy and F1 scores, while extreme structural filters like Sobel universally cause massive performance regressions. However, smaller anomalies exist, such as ResNet18 performing uncharacteristically poorly on the baseline (Acc: 65.20%) but jumping to 78.83% with a Sharpening filter.

5. Does filtering improve or decrease macro-F1 and balanced accuracy?

In the vast majority of cases, filtering decreases the macro-F1 score. Because macro-F1 treats all classes equally, it highlights a model's inability to classify minority classes. Filters obscure the subtle textures required to identify rare lesion types, forcing the models to default to predicting the dominant class ('Melanocytic nevi'), which severely tanks the macro-F1 score.

6. Which lesion classes are most affected by filtering?

Minority classes, such as Dermatofibroma, Actinic keratoses, and Vascular lesions, are disproportionately affected. The confusion matrix generated for the baseline ResNet18 model already shows zero accurate predictions for Vascular lesions and Dermatofibroma. Applying destructive filters like Sobel completely erases the nuanced features of these rare conditions, driving their individual F1 scores to near zero.

7. Why might smoothing remove useful lesion texture or morphological information?

Smoothing filters (like Average or Median) act as low-pass filters that average pixel intensities. Dermatological diagnoses heavily rely on high-frequency visual features,

such as sharp pigment networks, crisp borders, dots, and streaks. Smoothing blurs these sharp transitions into uniform patches, effectively destroying the primary mathematical variance the convolutional layers rely on to distinguish benign lesions from melanomas.

8. Why might sharpening or edge detection help or hurt classification?

Sharpening amplifies high-frequency details, which can sometimes help by making faint lesion borders more defined (as seen in the ResNet18 sharpening boost). However, it mostly hurts because it also amplifies meaningless background noise, hair, and skin texture, which the model may mistakenly learn as features. Edge detection (Sobel) fundamentally hurts skin-lesion classification because it entirely removes color data—one of the most critical diagnostic criteria in dermatology.

9. What is the difference between convolution and correlation?

In spatial domain image processing, correlation involves sliding a kernel across an image to compute a sum of products. Convolution is mathematically identical to correlation, with the key exception that the kernel is rotated by 180 degrees (flipped horizontally and vertically) before it is applied to the image.

10. Based on your results, explain the relationship between classical image processing and deep-learning-based feature extraction.

Classical image processing applies predefined, human-engineered kernels (like Gaussian or Sobel) to extract specific structural features. Deep learning, however, uses backpropagation to dynamically learn complex, hierarchical kernels optimized specifically for the training data. The experimental results demonstrate that applying rigid classical filters *before* deep learning restricts the model's potential. By artificially modifying the raw pixels with classical filters, we deprive the deep learning network of the complete RGB spatial data it needs to independently extract the most optimal features for classification.